

What we need to know.

- Go through quizzes!
- 50 marks.
- If in CA, not in exam, except 1 thing.
- Last 10 marks numbers
- First 40 0 numbers
- Chapter 1
 - ↳ concepts NB, Σ vector not tested directly but in context.
 - ↳ Notation is NB!
 - ↳ Write sample covariance matrix. (pg 4)
 - ↳ How can we relate Deviation vector to angle (pg 9) (point 3! w/ cosine) ^{angle.}
 - ↳ Must be able to derive covariances and write the different forms of sample cov matrix (quiz question)
 - ↳ Linear combos how to relate to X persons.

Chapter 2

- ↳ No SVD/eigenvalue/eigen vector calc.
- ↳ 2.3, show it's best low rank approx. (PROOF)

Chapter 3 (homework NB)

- ↳ First Q in exam give pdf of MVN.
- ↳ 3.3, from CA only exception. Show $(x-\mu)' \Sigma^{-1} (x-\mu) \sim \chi^2$ (Question 3.2)
 - ↳ Two ways (could ask both) $\rightarrow$ know the 2nd way. (one w/o eigen vectors with the $Y = \dots$)
- ↳ Know the properties of MVN (3.4 $\rightarrow$)
- ↳ Conditional (3.4.4) distribution proof, we are given the alt matrix, $A = \begin{bmatrix} I & -\Sigma_{12} \Sigma_{22}^{-1} \\ 0 & I \end{bmatrix}$
- ↳ 3.5 NB.
- ↳ 3.6 homework good to look at.

Chapter 4 (NB) this is a everything except. (Theorems A.1 and A.2 given to us)

- NO showing that our cov is indeed γ maximum. (Don't examine it) $\Rightarrow$ stop at $\frac{1}{n} A = \frac{n-1}{n} S$ (top of 3.7) bas
- ↳ improve to do substituting in rule (pg 38 is NB.)
- ↳ No Wishart pdf. (not so NB.) [must know sample cov matrix is Wishart dist]
- ↳ 4.2 is NB.
- NO 4.4.

Chapter 5

- ↳ 5.2 NB.
- ↳ 5.3 NB $\rightarrow$ must be able to reproduce. ^{1) define ratio ($\frac{\text{unrestricted}}{\text{restricted}}$)}
- ↳ Know how to do μ_0 at 35.
- ↳ Explain how we use Wilks lambda and link to Hotelling's T^2 .
- ↳ pg 51 NB
- ↳ Must be able to define MV confidence region top of (5.4).
- ↳ No point in Simultaneous CIs (Must be asked why we should be critical of them?) nothing else.

- G.L must look at. (top of pg 56 to example 4.6 by 4.7) NB.
- Repeated measures test as NB as done in CA.

MVA B

- MCQ questions 8 marks
- Out of 60.
- 3 data sets
- All content except some in DA.
- Priors are good guide.
- From scratch in firsts LDA will do! (6 marks to make from scratch)
 - PCA
 - most of the rest should be from first principals.
- interpretation matters!
- D0 Q1 + Q2 → for MCQs → ^{like to think about a MCQs} these

→ For DA

→ Code up LDA

1.) $\underline{a}' B \underline{a}$ s.t. $\underline{a}' S_{pooled}$

2.) $B = \sum_{i=1}^g n_i (\bar{x}_i - \bar{x})(\bar{x}_i - \bar{x})'$

3.) $S_{pooled}' B \underline{x} = \lambda \underline{x}$

→ $eigen(S_{pooled}' B)$

→ $\underline{a}' S_{pooled} = 1$ must hold

$\therefore \underline{e}_1' S_{pooled} \underline{e}_1 = 3$ for example

for we must divide left hand side by 3.